Conservation by radiation attempts to achieve the complete destruction of all bacteria with minimum change in the taste of the food (due to the formation of small amounts of decomposition products). Radiation pasteurization (i.e. partial sterilization with lower doses) and irradiation at low temperatures cause correspondingly smaller taste changes.

The safety of food irradiation has been accepted by IAEA, FAO (Food and Agricultural Organization) and WHO. It has been used for several decades without any negative health effects. More than 35 countries have approved some 50 different irradiated food products for unrestricted human consumption. Examples of the foods are (country, maximum dose): potatoes (Canada, 100 Gy), onions (Israel, 100 Gy), shrimp (Netherlands, 1000 Gy), fried meat in plastic bags (Soviet Union, 8000 Gy), and wheat and wheat products (USA, 500 Gy).

Medical supplies, which must be sterile, can be manufactured and packed by conventional techniques, after which the packages may be exposed to high energy penetrating radiation (e.g. ^{60}Co, ^{137}Cs or electrons from small accelerators). The radiation kills all bacteria with little damage to low molecular weight compounds. In this common technique, the irradiation source is quite large (0.1−1 MCi; around 10^{11}Bq). The packages slowly pass through the source, such that the doses are on the order of 10−30 kGy. The same facilities can be used also for other purposes such as food sterilization.

Radiation doses of ~2 Gy produce sterility in the human male for about 12 months while higher doses lead to permanent sterility; libido and potency are unchanged. Permanent sterility occurs in 60% of young women given 8 Gy of fractionated irradiation to the ovary, but occasional pregnancies can occur after doses of up to 5−8 Gy; the children are apparently normal at birth (BEIR V).

Radiation has been used to sterilize (80 Sv) the males of certain insect species which when released (the ratio between the sterilized and untreated males was approximately 4:1), mate with females and prevent further reproduction of the species. This technique has been used in the US, Mexico, Egypt, Libya, etc to eradicate screw worm flies which can cause huge damage to the cattle industry. The technique is now also used against other insect species which are threats to agriculture.

15.8. GENETIC EFFECTS

In 1927, Muller showed how irradiation of fruit flies (Drosophila M.) could produce new species that were defective with regard to the parents (e.g. lacked wings). This defect was carried in the genes to later generations. This dramatic demonstration of the mutagenic effect of radiation has been extended to other primitive species. The vast majority of such changes are recessive, i.e. in order to become dominant both parents must carry the same DNA-damage. From all experience as well as the theoretical considerations described above, the likelihood for two identical DNA impairments to

come together in the fertilized cell is extremely small. No radiation induced hereditary effects have been observed in mammals. Irradiation of the sexual organs of 2000 carefully selected mice (~2 Sv each) for 19 generations (in total 38 000 mice, all inbred) showed no genetic changes (Lüning 1970). A thorough investigation of the 75 000 children born to parents who were exposed to the A-bomb irradiation in 1945 have not shown any increase in the frequency of hereditary diseases (or cancer). The 35 000 children born to parents of which at least one had been exposed to $\leq$3 Sv (average exposure 0.2 Sv) showed no genetic differences (analyzing still births, birth weight, congenital abnormalities, infant mortality, childhood mortality, leukemia or sex ratio) from children born to non-irradiated parents within statistical uncertainty.

Large doses to plants have caused mutations which have either improved the quality of the species or produced effects which are disadvantageous to the species, but desirable to society. Although irradiation of plant seeds results in a ratio of about 1000 to 1 of the harmful to the advantageous mutations, by cultivating those few plants showing improvement in properties, new plant variations have been obtained. This has resulted in species of grains and legumes which have stronger stocks, higher yields, and improved resistance to mold and to fungi. In northern countries such as Sweden, most of the grain that is grown today is radiation-produced species possessing much greater cold resistance.

As a conclusion one might say that genetic effects in higher mammals from radiation has not been observed. After the Chernobyl accident thorough studies of fast reproducing mammals such as rats have shown no genetic effects.

15.9. RADIOMIMETIC SUBSTANCES

Many chemical substances when administrated to the body show the same effects as irradiation. Because cancer caused by radiation has been investigated more than cancer induced by chemicals, such substances are called *radiomimetic* (mimetic = imitative). In order to qualify as a radiomimetic agent, the substance must do the following: stop cell division, stop tumor formation, produce chromosome aberrations, cause mutations, kill lymphocytes, and be carcinogenic. Chemical substances which meet a few, but not all of these requirements are not considered radiomimetic. The effects depend on the concentration of the substance; e.g. cell division is interrupted by many radiomimetic substances at concentrations $\leq 10^{-5}$ mole l^{-1}.

Typical radiomimetic substances are organic peroxides (e.g. ethylene oxide), ethylene di-imine, mustard gas and derivatives, aliphatic dichloro-amines, etc. These compounds or chemical groups occur in many familiar materials such as tobacco smoke. The effect of a certain amount of a radiomimetic substance can be calculated to correspond to a radiation dose. Thus, smoking a pipe of tobacco corresponds to an average radiation dose of about 0.04 mSv.